

IT ISN'T A bird. It isn't a plane. And no, it isn't Superman either. Neither is it an alien object. What, then, is this flying machine that has everyone from bloggers to founders to scientists excited? It is a drone. Weighing between 20 gm and 100 kg, drones are touted as the next frontier in technology—along with IoT devices, AI and ML—there is a use-case for drones in practically every sector.

And naturally, business is beginning to boom. From just a handful of drone companies till a decade ago, India is today believed to have up to 250 such firms across manufacturing and services. “The indigenous

drone manufacturing ecosystem has changed tremendously in the past few years,” says Neel Mehta, Director and Co-founder of Bengaluru-based full-stack drone tech firm Asteria Aerospace, a subsidiary of Reliance's Jio Platforms. “By creating an enabling regulatory framework and policy, the government is encouraging manufacturers to enhance capabilities.” The introduction of the liberalised Drone Rules, 2021, paved the way for innovation throughout the sector. “As drone imports have been banned, there has been a surplus demand for Make in India drones in the country, leading to tremendous scope for growth in the manufacturing sector,” says Prateek Srivastava, Founder & MD of DroneAcharya Aerial Innovations, a Pune-based provider of drone and GIS technology courses and solutions.

However, he warns that domestic manufacturers need to come up to the level of quality set by global pioneers in the field so that they can meet the requirements of customers who were buying imported drones earlier. But make no mistake, quality levels are not the only mountains to be climbed. There are many more challenges the nascent industry faces, but more on that later.

THE TRIGGER

Let us first go back a few years to track the growth in the sector. While the pandemic-induced lockdowns in 2020 caused a slowdown across several industrial segments, drones got a boost. India did announce a major policy in

2020

THE SKY IS THE LIMIT

India has the potential to become a drone manufacturing hub

Total market size of India's drone industry (commercial and defence combined):
₹2,900 crore

2025*

Counter-unmanned aircraft system:
₹20,000 crore

Defence: **₹38,300 crore**

Commercial: **₹13,000 crore**

Homeland security: **₹6,000 crore**

Total market size: **₹77,300 crore**

40 per cent hardware indigenisation:
₹32,640 crore

Exports: **₹4,300 crore**

2030*

Counter-unmanned aircraft system: **₹49,000 crore**

Defence: **₹1,01,000 crore**

Commercial: **₹75,000 crore**

Homeland security: **₹23,000 crore**

Total market size: **₹2,48,000 crore**

60 per cent hardware indigenisation:
₹1,77,000 crore

Exports:
₹47,000 crore

*ESTIMATES; **SOURCE** EY-FICCI'S 'MAKING INDIA THE DRONE HUB OF THE WORLD' REPORT



BATTLE READY

Challenges that India's developing drone ecosystem needs to address



Training: The country requires more educational institutes that offer degree courses on drones



Trackers: Stakeholders have failed to come up with a solution to the menace of unauthorised drones



Supply-side issues: Players need to standardise components and then mass produce them



Counter UAS: Proper defensive strategies are needed to prevent cross-border drone infiltration

SOURCE INDUSTRY, BT RESEARCH